

MODEL QUESTION PAPER 1**S5 INSTRUMENTATION ENGINEERING****COURSE: INDUSTRIAL INSTRUMENTATION***Time: 3 Hour**Max. Marks: 75***I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)**

		Module Outcomes	Cognitive Level
1	Define the term pH.	M1.01	U
2	List one instrument used for viscosity measurement.	M1.02	R
3	Define ‘absolute humidity’	M2.01	U
4	Name two instruments used for nuclear radiation measurement.	M2.03	R
5	What is the use of stroboscope?	M3.03	U
6	Name one device used for force measurement.	M3.02	R
7	What is the basic principle of thickness measurement using ultrasonic method ?	M4.02	U
8	Expand the term MEMS.	M4.03	R
9	What is the use of hygrometers ?	M2.01	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		Module Outcomes	Cognitive Level
1.	Describe the terms ‘absolute viscosity’, ‘kinematic viscosity’ and ‘relative viscosity’.	M1.01	U
2.	Draw the picture of glass electrode.	M1.03	U
3.	Illustrate the working of hair hygrometer.	M2.02	U
4.	Define the terms ‘relative humidity’ and ‘absolute humidity’.	M2.01	U
5.	Illustrate any one type of torque measurement.	M3.02	U
6.	Describe any one type of revolution counter.	M3.03	U
7.	Illustrate inductive type thickness measurement.	M4.02	U
8.	Write short notes on MEMS.	M4.03	U
9.	Describe working of AC tachogenerator.	M3.02	U
10.	Describe the working of capacitance type hygrometer.	M2.02	U

Part – C

Answer any six questions. Each question carries 7 marks. (7 × 6 = 42 Marks)

		Module Outcomes	Cognitive Level
III	With necessary figures explain pH measurement using combined electrode.	M1.03	U
	OR		
IV	With necessary figures explain density measurement using LVDT	M1.02	U
V	With necessary figures explain construction and working of wet bulb psychrometer	M2.02	U
	OR		
VI	With necessary figures explain construction and working of Gieger Muller tube.	M2.03	U
VII	With necessary figures explain force measurement using pneumatic load cell.	M3.02	U
	OR		
VIII	With necessary figures explain any one method of speed measurement	M3.03	U
IX	Explain acceleration measurement using piezoelectric accelerometer.	M4.02	U
	OR		
X	Explain any one application of nano sensors.	M4.03	U
XI	With necessary figures explain working of eddy current proximity sensor.	M4.01	U
	OR		
XII	Explain any one type of smoke detector.	M4.03	U
XIII	With necessary figures explain digital pH meter.	M1.03	U
	OR		
XIV	With necessary figures explain specific gravity measurement.	M1.02	U

MODEL QUESTION PAPER 2**S5 INSTRUMENTATION ENGINEERING****Course: INDUSTRIAL INSTRUMENTATION***Time: 3 Hour**Max. Marks: 75***I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)**

		Module Outcomes	Cognitive Level
1	Write one unit of viscosity.	M1.01	R
2	Define specific gravity.	M1.01	U
3	What is dew point?	M2.01	U
4	Name two instruments used for nuclear radiation measurement.	M2.03	R
5	What is the use of load cell?	M3.02	R
6	Name one device used for speed measurement.	M3.02	U
7	What is the basic principle of thickness measurement using capacitive method?	M4.02	U
8	Expand the term MEMS.	M4.03	R
9	Write applications of radiation detector.	M1.03	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		Module Outcomes	Cognitive Level
1.	Describe the terms 'absolute viscosity', 'kinematic viscosity' and 'relative viscosity'.	M1.01	U
2.	Draw the picture of hydrometer.	M1.02	U
3.	Illustrate the working of resistive hygrometer	M2.02	U
4.	What are the applications of radiation sensors?	M2.03	U
5.	Illustrate any one type of torque measurement.	M3.02	U
6.	Describe eddy current tachometer.	M3.03	U
7.	Compare relative humidity and absolute humidity.	M2.01	U
8.	Write short notes on Nano sensors.	M4.03	U
9.	Describe working of stroboscope.	M3.02	U
10.	Write short notes on mechanical vibration sensor	M4.01	U

Part – C

Answer any six questions. Each question carries 7 marks. (7 × 6 = 42 Marks)

		Module Outcomes	Cognitive Level
III	Draw and explain Redwood Viscometer.	M1.02	U
	OR		
IV	Describe any two electrodes used for pH measurement.	M1.03	U
V	With necessary figures explain wet and dry bulb hygrometer	M2.02	U
	OR		
VI	With necessary figures explain construction and working of Ionization chamber.	M2.03	U
VII	With necessary figures explain force measurement using hydraulic load cell.	M3.02	U
	OR		
VIII	With necessary figures explain the working of proximity torque sensor	M3.03	U
IX	Explain acceleration measurement using LVDT accelerometer.	M4.02	U
	OR		
X	Describe any one type of smoke detector.	M4.03	U
XI	Describe the working of inductive methods of thickness measurement.	M4.01	U
	OR		
XII	Draw and explain seismic accelerometer.	M4.02	U
XIII	With necessary figures explain digital pH meter.	M1.03	U
	OR		
XIV	Describe the working of saybolt viscometer with figure.	M1.02	U